

Yara Naif Alotaibi

Applied AI Engineer | LLM and Generative AI | End-to-End AI Systems
Taif, Saudi Arabia | +966 547253104 | yaranaifalotaibi@gmail.com
www.linkedin.com/in/yara-alotaibi-b22094300 | https://github.com/YaraNaif

PROFILE SUMMARY

Applied AI Engineer with hands-on experience designing, building, and deploying end-to-end AI systems spanning machine learning, deep learning, and Large Language Models (LLMs). Built and deployed a production-oriented bilingual RAG assistant (Qwen2.5-7B, FAISS, Docker) and an IEEE-published seizure prediction model (99.49% accuracy, AUC = 1.00), combining research rigor with practical deployment across the full model lifecycle. AiE®-certified, with hands-on MLOps experience using FastAPI, Azure, MLflow, and DVC.

EDUCATION

BSc in Computer Science – Umm Al-Qura University, Makkah | Aug 2021 - Jun 2025

Generative AI (MLOps Specialization) Bootcamp – Saudi Digital Academy and WeCloudData, Remote | Aug 2025 - Nov 2025

- Industry-focused specialization covering the end-to-end ML lifecycle, cloud deployment (Docker, Azure), and production LLM systems (MLflow, DVC, FastAPI).

WORK EXPERIENCE

AI Engineering Consultant – Industry Capstone, Beam Data and Saudi Digital Academy, Remote | Aug 2025 – Nov 2025

As part of a 12-week industry-aligned Generative AI and MLOps program, delivered production-oriented AI systems under real industry mentorship:

- Designed and built end-to-end LLM applications, including RAG pipelines, prompt engineering workflows, and automated evaluation frameworks.
- Developed scalable AI services using FastAPI, Docker, and Azure, enabling deployment-ready inference through containerized microservices.
- Applied MLOps practices using MLflow and DVC for experiment tracking, model versioning, and reproducible ML workflows.
- Evaluated LLM performance using retrieval and generation metrics, iteratively reducing hallucinations and improving response quality.
- Collaborated and communicated within a cross-functional engineering team to deliver production-ready prototypes evaluated by Beam Data leadership.

Summer Internship – Ministry of Education, Taif | Jun 2024 – Aug 2024

- Developed a secure, role-based web application using Laravel and MySQL, analyzing 300+ school data records using Excel and Access.

PROJECTS

SALAM AI Legal Assistant for Saudi Labor Law – Generative AI Bootcamp Capstone (Beam Data and Saudi Digital Academy), Remote | Aug 2025 - Nov 2025 | [Link](#)

Designed and developed an end-to-end bilingual AI legal assistant enabling users to retrieve and understand Saudi Labor Law using LLMs and Retrieval-Augmented Generation (RAG).

- Engineered a complete RAG pipeline using Qwen2.5-7B, FAISS, FastAPI, and Docker to deliver grounded legal responses in Arabic and English.
- Achieved 95.65% retrieval recall through Cross-Encoder reranking, with 83.02% F1 score and 73.33% precision on evaluation benchmarks.
- Reduced LLM hallucinations via grounded retrieval with legal citation 70% of responses correctly cite the relevant Saudi Labor Law article.
- Deployed a fully Dockerized microservices architecture separating API, embedding generation, and vector search for scalability.

Seizable AI System for Epileptic Seizure Prediction – Umm Al-Qura University, Makkah | May 2025

Led the design and development of an end-to-end AI system for epileptic seizure prediction using multichannel EEG signals, combining deep learning, signal processing, and web deployment.

- Structured the project into specialized workstreams data preprocessing, hyperparameter optimization, results benchmarking, overfitting validation, and backend/web development and led a team of student engineers by assigning ownership, setting milestones, and integrating outputs across the pipeline.
- Independently engineered the signal-processing core of the pipeline converting raw EEG ADU values to voltage and frequency domain, and building the visualization layer for clinical-style EEG waveforms (Plotly).
- Designed and optimized a hybrid CNN–TCN architecture across multiple experiments, achieving 99.49% accuracy and AUC = 1.00 (10-fold cross-validation).
- Developed an interactive prediction platform integrating Flask, MySQL, and deep learning inference.
- Awarded 1st Place, UQU Entrepreneurial Projects Exhibition 2025; published as a peer-reviewed IEEE paper.

PUBLICATIONS

Hybrid CNN-TCN Modeling of EEG Temporal Signals for Epileptic Seizure Prediction

IEEE AMLDS 2025, Tokyo | Jul 2025 | [Link](#)

- Sole presenter and first author; authored the full methodology and mathematical formulation of EEG signal properties.
- Demonstrated statistically robust performance improvement over CNN, LSTM, and LSTM-TCN baselines across intra- and inter-patient EEG analysis.

TECHNICAL SKILLS

Artificial Intelligence: Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), NLP, Time-Series Analysis, Predictive Modeling

Frameworks and Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, FastAPI, Flask

MLOps and Deployment: Docker, MLflow, DVC, CI/CD, GitHub Actions, Model Deployment, Experiment Tracking

Cloud and Infrastructure: Microsoft Azure, REST APIs, Vector Databases (FAISS)

Programming: Python, SQL, PHP, JavaScript

Tools: Git, GitHub, VS Code, MySQL

LICENSES AND CERTIFICATIONS

Artificial Intelligence Engineer (AiE®) – Artificial Intelligence Board of America, Jun 2026 – Jun 2029 | [Link](#)

- Credential ID: 9609480435
- Aligned with the AMDEX™ Framework, validating applied expertise across the full AI/ML model lifecycle.

KAUST Academy Intro and Advanced AI Training, Makkah | Jan 2025 – Feb 2025

- Intensive AI program covering CNNs (ResNet, AlexNet), YOLO, SSD, and U-Net segmentation PyTorch.

Tuwaiq Academy Full-Stack Web Development, Remote | Mar 2024 – Apr 2024

- Built full-stack web applications using Laravel, MySQL, and XAMPP, applying MVC architecture, middleware, and Blade templating.

LANGUAGES

Arabic: Native speaker

English: Professional working proficiency
